

① Check:

beq \$a0, \$a1, else-case

if-case:

li \$v0, 1

j exit-check

else-case:

li \$v0, 0

exit-check:

jr \$ra

②

li \$s0, 0

li \$t0, 0

Loop:

bge \$t0, \$s1, endloop

sll \$t2, \$t0, 2

add \$t2, \$t2, \$s2

lw \$t3, 0(\$t2)

add \$s0, \$s0, \$t3

addi \$t0, \$t0, 1

j Loop

endloop:

③ a) Instruções estáticas: 8 (li, li, bge, sll, add, lw, addl, addi, j), se contar apenas o loop são 7.

b) Instruções dinâmicas: 59 instruções dinâmicas

$$④ \quad g = w - A[i]$$

⑤

```
addi $t0, $s1, -2
sll $t0, $t0, 2
add $t0, $t0, $s3
lw $t1, 0($t0)
addi $t1, $t1, 5
sll $t2, $s0, 2
add $t2, $t2, $s2
sw $t1, 0($t2)
```

⑥

```
int i;
for (i = 0; i < 100; i++) {
    result += memarray[i];
}
```

⑦

```
for (i = n; i > 0; i--) {
    Values[i] = Values[i] * 2;
}
```

⑧

sll \$t0, \$s0, 2	sll \$t0, \$s0, 2
add \$t0, \$t0, \$s2	add \$t0, \$t0, \$s4
lw \$t1, 0(\$t0)	sw \$t1, 0(\$t0)
sll \$t2, \$s1, 2	
add \$t2, \$t2, \$s3	
lw \$t3, 0(\$t2)	
add \$t1, \$t1, \$t3	

$$⑨ \quad g = A[i+4] - h;$$

$$⑩ \quad i = -1;$$

do {

$i++;$

} while ($A[i] \neq 0$);

⑪

li \$x0, 0

Loop:

orge \$x0, \$x1, EndLoop

sub \$t0, \$x1, \$x0

addi \$t0, \$t0, -1

sll \$t1, \$t0, 2

add \$t1, \$t1, \$x2

sw \$t2, 0(\$t1)

sll \$t3, \$x0, 2

add \$t3, \$t3, \$x3

sw \$t2, 0(\$t3)

addi \$x0, \$x0, 1

j Loop

EndLoop:

⑫ ser resolução do professor

